BABU BANARASI DAS UNIVERSITY

CASE STUDY PROJECT ON

"E-COMMERCE SALES DATA ANALYSIS"

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PROJECT

E-COMMERCE SALES DATA ANALYSIS

ABOUT:

-This project focuses on analyzing e-commerce sales data using the Hadoop ecosystem. The dataset contains information such as Order ID, Product Name, Category, Price, Quantity, and Date of purchase.

-The main objective of this project is to extract useful insights from the data, such as identifying top-selling products, calculating total sales revenue, and analyzing category-wise performance.

-The project uses distributed data processing techniques to efficiently handle and analyze large datasets.

ARCHITECTURE:

~The architecture of this project is based on the Hadoop framework, which consists of **three main components:**

~**HDFS (Hadoop Distributed File System):** Used to store the dataset in a distributed manner.

~**MapReduce:** Used to process the data and perform operations like counting and aggregation.

~**Hive:** Used to perform SQL-based queries for analysis.

DATA FLOW:

Raw Data (CSV File)

→ Uploaded to HDFS (/input)

→ Processed using MapReduce (mapper & reducer)

→ Output stored in HDFS (/output)

→ Data loaded into Hive table

→ Queries executed for analysis

→ Final insights generated

EXPLANATION:

~First, the dataset is uploaded into HDFS, which acts as a storage layer. Then, MapReduce is applied to process the data by counting the number of occurrences of each product. The mapper extracts the product name and emits key-value pairs, while the reducer aggregates the total count for each product.

~After processing, the results are stored in HDFS and further analyzed using Hive. Hive allows writing SQL-like queries to extract meaningful insights such as total revenue, category-wise sales, and peak sales dates.

~This combination of HDFS, MapReduce, and Hive helps in efficient big data processing and analysis.

TOOLS USED:

- Hadoop (HDFS , MAPREDUCE)
- Hive
- Cloudera

TECHNOLOGIES USED:

- Python (for mapreduce scripts)
- Linux (command line operations)
- Hive (SQL)

Use of Generative AI in the Project

Generative AI tools were used to assist in developing and completing the project. AI helped in generating code, fixing errors, and understanding Hadoop concepts.

Areas Where AI Was Used

-Code Generation: Used to create MapReduce (mapper and reducer) programs.

-Debugging: Helped in resolving errors like syntax issues and execution problems.

-Query Writing: Assisted in writing Hive queries such as GROUP BY and aggregation functions.

Benefits of Using AI

-Faster Development: Reduced time required to complete the project.

-Error Reduction: Helped in identifying and fixing mistakes quickly.

-Better Logic Suggestions: Improved the accuracy of code and queries.

-Easy Understanding: Simplified complex concepts like Hadoop, MapReduce, and Hive.

-Time Saving: Reduced manual effort in writing and testing code.

-Improved Productivity: Helped complete tasks efficiently within less time.

Dataset Description

This project uses an e-commerce sales dataset containing product details like price, quantity, and purchase date to analyze sales performance.

Dataset Structure

-The dataset is in CSV format and contains the following fields:

-OrderID: Unique identifier for each order

-Product: Name of the product

-Category: Product category (Electronics, Fashion, Accessories, etc.)

-Price: Price of the product

-Quantity: Number of units sold

-Date: Date of purchase

Dataset Source

The dataset used in this project is a sample/synthetic dataset created for educational purposes. It is designed to simulate real-world e-commerce sales data for analysis using big data tools.

Purpose of Dataset

The main purpose of this dataset is to perform analysis on sales data and gain insights such as:

-Identifying top-selling products

-Calculating total sales revenue

-Analyzing category-wise performance

-Finding peak sales dates

Data Validation

Basic checks were done for missing values, correct format, and valid numeric data.

PROBLEM STATEMENT :

To analyze e-commerce sales data using Hadoop, MapReduce, and Hive to find top products, total sales, and peak sales time.

STEP 1:

Creating Dataset in csv format or uploading a dataset.

-(nano ecommerce.csv)

-Upload the data and save the file.

[Dataset was created in CSV format containing product details like price, quantity, and date.]

```
OrderID,Product,Category,Price,Quantity,Date
1,Mobile,Electronics,10000,2,2024-01-01
2,Laptop,Electronics,50000,1,2024-01-02
3,Shoes,Fashion,2000,3,2024-01-03
4,Tshirt,Fashion,800,5,2024-01-04
5,Headphones,Electronics,1500,4,2024-01-05
6,Watch,Accessories,3000,2,2024-01-06
7,Bag,Accessories,1200,3,2024-01-07
8,Tablet,Electronics,20000,1,2024-01-08
9,Jeans,Fashion,2500,2,2024-01-09
10,Charger,Electronics,500,6,2024-01-10
11,Mobile,Electronics,10000,1,2024-01-11
12,Laptop,Electronics,50000,2,2024-01-12
13,Shoes,Fashion,2000,1,2024-01-13
14,Tshirt,Fashion,800,3,2024-01-14
15,Headphones,Electronics,1500,2,2024-01-15
16,Watch,Accessories,3000,4,2024-01-16
17,Bag,Accessories,1200,2,2024-01-17
18,Tablet,Electronics,20000,1,2024-01-18
19,Jeans,Fashion,2500,3,2024-01-19
20,Charger,Electronics,500,5,2024-01-20
21,Mobile,Electronics,10000,3,2024-01-21
22,Laptop,Electronics,50000,1,2024-01-22
23,Shoes,Fashion,2000,2,2024-01-23
24,Tshirt,Fashion,800,4,2024-01-24
25,Headphones,Electronics,1500,3,2024-01-25
26,Watch,Accessories,3000,2,2024-01-26
27,Bag,Accessories,1200,1,2024-01-27
28,Tablet,Electronics,20000,2,2024-01-28
29,Jeans,Fashion,2500,2,2024-01-29
30,Charger,Electronics,500,4,2024-01-30
31,Mobile,Electronics,10000,2,2024-02-01
32,Laptop,Electronics,50000,1,2024-02-02
33,Shoes,Fashion,2000,3,2024-02-03
34,Tshirt,Fashion,800,5,2024-02-04
35,Headphones,Electronics,1500,4,2024-02-05
36,Watch,Accessories,3000,2,2024-02-06
37,Bag,Accessories,1200,3,2024-02-07
38,Tablet,Electronics,20000,1,2024-02-08
39,Jeans,Fashion,2500,2,2024-02-09
40,Charger,Electronics,500,6,2024-02-10
41,Mobile,Electronics,10000,1,2024-02-11
```

STEP 2:

-Create hdfs directory

(hdfs dfs -mkdir /input)

STEP 3:

-Uploading dataset to hdfs

(hdfs dfs -put ecommerce.csv /input)

The dataset file was uploaded into HDFS using put command.

```
[cloudera@quickstart Desktop]$ hdfs dfs -put ecommerce.csv /input
[cloudera@quickstart Desktop]$ hdfs dfs -ls /input
Found 1 items
-rw-r--r--   1 cloudera supergroup       3959 2026-04-22 22:49 /input/ecommerce.csv
[cloudera@quickstart Desktop]$
```

MapReduce is used to process the dataset and count the number of occurrences of each product (i.e., find top-selling products).

-<u>Creating Mapper program so as to extract product names and assign value 1.</u>

(nano mapper.py)

-Hence inserting the following code:

```
import sys
for line in sys.stdin:
    data = line.strip().split(",")
    if len(data) > 1:
        product = data[1]
        print product + "\t1"
```

-<u>Creating reducer program so as to count total occurences of each product.</u>

(nano reducer.py)

-Hence inserting the following reducer code:

```
import sys
current_product = None
count = 0

for line in sys.stdin:
    line = line.strip()
    product, value = line.split("\t")

    if current_product == product:
        count += int(value)
    else:
        if current_product:
            print current_product, count
        current_product = product
        count = int(value)

if current_product:
    print current_product, count
```

-Now running MAPREDUCE JOB:

(Map reduce job was executed using Hadoop streaming.)

```
[cloudera@quickstart Desktop]$ hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar \
> -file mapper.py \
> -file reducer.py \
> -input /input/ecommerce.csv \
> -output /output \
> -mapper "python mapper.py" \
> -reducer "python reducer.py"
26/04/22 23:20:09 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [mapper.py, reducer.py] [/usr/jars/hadoop-streaming-2.6.0-cdh5.4.2.jar] /tmp/streamjob8456843951525095172.jar tmpDir=null
26/04/22 23:20:10 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/04/22 23:20:10 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/04/22 23:20:11 INFO mapred.FileInputFormat: Total input paths to process : 1
26/04/22 23:20:11 INFO mapreduce.JobSubmitter: number of splits:2
26/04/22 23:20:11 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1776920535638_0003
26/04/22 23:20:12 INFO impl.YarnClientImpl: Submitted application application_1776920535638_0003
26/04/22 23:20:12 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1776920535638_0003/
26/04/22 23:20:12 INFO mapreduce.Job: Running job: job_1776920535638_0003
26/04/22 23:20:17 INFO mapreduce.Job: Job job_1776920535638_0003 running in uber mode : false
26/04/22 23:20:17 INFO mapreduce.Job:  map 0% reduce 0%
26/04/22 23:20:22 INFO mapreduce.Job:  map 50% reduce 0%
26/04/22 23:20:23 INFO mapreduce.Job:  map 100% reduce 0%
26/04/22 23:20:27 INFO mapreduce.Job:  map 100% reduce 100%
26/04/22 23:20:27 INFO mapreduce.Job: Job job_1776920535638_0003 completed successfully
26/04/22 23:20:27 INFO mapreduce.Job: Counters: 49
        File System Counters
                FILE: Number of bytes read=1119
                FILE: Number of bytes written=343979
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=6145
                HDFS: Number of bytes written=120
                HDFS: Number of read operations=9
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
        Job Counters
                Launched map tasks=2
                Launched reduce tasks=1
                Data-local map tasks=2
                Total time spent by all maps in occupied slots (ms)=5102
                Total time spent by all reduces in occupied slots (ms)=2489
                Total time spent by all map tasks (ms)=5102
                Total time spent by all reduce tasks (ms)=2489
                Total vcore seconds taken by all map tasks=5102
```

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Now to view the output :

(hdfs dfs -cat /output/part-00000)

-The processed output was as follows:

```
                Bytes Written=120
26/04/22 23:20:27 INFO streaming.StreamJob: Output directory: /output
[cloudera@quickstart Desktop]$ hdfs dfs -ls /output
Found 2 items
-rw-r--r--   1 cloudera supergroup          0 2026-04-22 23:20 /output/_SUCCESS
-rw-r--r--   1 cloudera supergroup        120 2026-04-22 23:20 /output/part-00000
[cloudera@quickstart Desktop]$ hdfs dfs -cat /output/part-00000
Bag 10
Charger 10
Headphones 10
Jeans 10
Laptop 10
Mobile 11
Product 1
Shoes 10
Tablet 10
Tshirt 10
Watch 10
[cloudera@quickstart Desktop]$
```

<u>Hive was started to perform SQL – Based analysis:</u>

Hive is used to perform SQL-based analysis on the dataset to find insights like total sales, category-wise performance, and peak sales.

STEP 1:

Creating table in hive to store the dataset structure:

STEP 2:

Loading the dataset and performing queries:

LOAD DATA INPATH '/input/ecommerce.csv' INTO TABLE sales;

```
[cloudera@quickstart Desktop]$ hive

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j.properties
WARNING: Hive CLI is deprecated and migration to Beeline is recommended.
hive> CREATE TABLE sales (
    > OrderID INT,
    > Product STRING,
    > Category STRING,
    > Price INT,
    > Quantity INT,
    > Date STRING
    > )
    > ROW FORMAT DELIMITED
    > FIELDS TERMINATED BY ',';
OK
Time taken: 1.876 seconds
hive> LOAD DATA INPATH '/input/ecommerce.csv' INTO TABLE sales;
Loading data to table default.sales
chgrp: changing ownership of 'hdfs://quickstart.cloudera:8020/user/hive/warehouse/sales/ecommerce.csv': User does not belong to hive
Table default.sales stats: [numFiles=1, totalSize=3959]
OK
Time taken: 0.622 seconds
hive>
```

Query 1: Top Products

This query calculates total quantity sold for each product.

```
hive> SELECT Product, SUM(Quantity) AS total_sold
    > FROM sales
    > GROUP BY Product
    > ORDER BY total_sold DESC;
Query ID = cloudera_20260423003030_bf735f28-2d93-4b72-b4e7-733d1958789e
Total jobs = 2
Launching Job 1 out of 2
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 2.91 sec   HDFS Read: 10367 HDFS Write: 387 SUCCESS
Stage-Stage-2: Map: 1  Reduce: 1   Cumulative CPU: 2.26 sec   HDFS Read: 4761 HDFS Write: 116 SUCCESS
Total MapReduce CPU Time Spent: 5 seconds 170 msec
OK
Charger 52
Tshirt  40
Headphones      30
Watch   28
Jeans   24
Bag     22
Shoes   20
Mobile  19
Laptop  14
Tablet  12
Product NULL
NULL    NULL
Time taken: 43.541 seconds, Fetched: 12 row(s)
hive>
```

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Query 2: Total Revenue

This query calculates total sales revenue.

```
ive> SELECT SUM(Price * Quantity) AS total_revenue
   > FROM sales;
Query ID = cloudera_20260423003333_771856d8-6bbb-47fc-a4db-341e7ee3aaf6
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1776920535638_0006, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1776920535638_0006/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1776920535638_0006
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-04-23 00:33:34,539 Stage-1 map = 0%,  reduce = 0%
2026-04-23 00:33:40,799 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 1.49 sec
2026-04-23 00:33:47,140 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 2.83 sec
MapReduce Total cumulative CPU time: 2 seconds 830 msec
Ended Job = job_1776920535638_0006
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 2.83 sec   HDFS Read: 11972 HDFS Write: 8 SUCCESS
Total MapReduce CPU Time Spent: 2 seconds 830 msec
OK
1443400
Time taken: 20.577 seconds, Fetched: 1 row(s)
ive>
```

Query 3: Category-wise Analysis

This query shows sales performance for each category.

```
hive> SELECT Category, SUM(Quantity) AS total_items
    > FROM sales
    > GROUP BY Category;
Query ID = cloudera_20260423003434_4394048a-7239-4ec2-a8e7-a8d1eb824127
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1776920535638_0007, Tracking URL = http://quickstart.cloudera:8088/pr
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1776920535638_0007
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-04-23 00:34:51,008 Stage-1 map = 0%,  reduce = 0%
2026-04-23 00:34:56,245 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 0.84 sec
2026-04-23 00:35:02,517 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 2.21 sec
MapReduce Total cumulative CPU time: 2 seconds 210 msec
Ended Job = job_1776920535638_0007
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 2.21 sec   HDFS Read: 11114 HDFS Writ
Total MapReduce CPU Time Spent: 2 seconds 210 msec
OK
NULL    NULL
Accessories     50
Category        NULL
Electronics     127
Fashion 84
Time taken: 18.965 seconds, Fetched: 5 row(s)
hive>
```

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QUERY 4: Peak sales date.

This query is used to identify the date on which the highest sales were recorded (peak sales time).

```
hive> SELECT Date, SUM(Quantity) AS total_sales
    > FROM sales
    > GROUP BY Date
    > ORDER BY total_sales DESC
    > LIMIT 1;
Query ID = cloudera_20260423003737_40f1bcf4-bfd6-4bdd-923c-b987a8e372ba
Total jobs = 2
Launching Job 1 out of 2
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1776920535638_0008, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1776920535638_0008/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1776920535638_0008
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-04-23 00:37:42,767 Stage-1 map = 0%,  reduce = 0%
2026-04-23 00:37:47,944 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 0.89 sec
2026-04-23 00:37:54,145 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 2.23 sec
MapReduce Total cumulative CPU time: 2 seconds 230 msec
Ended Job = job_1776920535638_0008
Launching Job 2 out of 2
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1776920535638_0009, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1776920535638_0009/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1776920535638_0009
Hadoop job information for Stage-2: number of mappers: 1; number of reducers: 1
2026-04-23 00:38:01,957 Stage-2 map = 0%,  reduce = 0%
2026-04-23 00:38:07,158 Stage-2 map = 100%,  reduce = 0%, Cumulative CPU 0.89 sec
2026-04-23 00:38:13,559 Stage-2 map = 100%,  reduce = 100%, Cumulative CPU 2.24 sec
MapReduce Total cumulative CPU time: 2 seconds 240 msec
Ended Job = job_1776920535638_0009
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 2.23 sec   HDFS Read: 10388 HDFS Write: 3055 SUCCESS
Stage-Stage-2: Map: 1  Reduce: 1   Cumulative CPU: 2.24 sec   HDFS Read: 7547 HDFS Write: 13 SUCCESS
Total MapReduce CPU Time Spent: 4 seconds 470 msec
OK
2024-01-10      6
Time taken: 38.083 seconds, Fetched: 1 row(s)
hive>
```

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CONCLUSION:

-The project successfully demonstrated the use of Hadoop ecosystem tools for analyzing e-commerce sales data. Data was stored in HDFS, processed using MapReduce, and analyzed using Hive queries.

-The analysis helped in identifying top-selling products, calculating total revenue, and understanding category-wise performance. This project shows how big data technologies can be used to efficiently handle and analyze large datasets.